# Microbe isolation: environmental sample → primary plate → pure culture → preservation vials

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v2.1, 2026-10-20; data steps via /lims-register (Stage 1 → 2a → 2b → Culture#)

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**Source:** rewritten from the lab's protocols.io protocol *Step by Step from Environmental Samples to Preservation Vials* (dx.doi.org/10.17504/protocols.io.n2bvj8mk5gk5/v1). Bench steps unchanged; the Specify datasheet upload is replaced by `/lims-register`. Do not modify the bench steps without telling the lab manager.

**Route:** environmental sample (Stage 1, Specimen#) → primary plate (Stage 2a) → one secondary plate per isolate (Stage 2b, Culture#) → preservation vials + a lysis tube for DNA extraction (Stage 3a).

**Procedure**

### STAGE 1 · REGISTER THE SPECIMENS

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**1.** Catalogue every container that arrived as **one Stage-1 specimen** in LIMS: run `/lims-register` with your sample list (spreadsheet, CSV or a photo of the labelled bags/tubes). Method = `Soil`, `Fungal strain`, etc.; collector, date, locality as known. Confirm the preview; LIMS mints the **Specimen#** and prints the labels.

**Note** A strain collection (many isolates from one source) is registered as **one umbrella Stage-1 specimen**; do not split by collection event.

### STAGE 2A · ISOLATE PRIMARY CULTURES

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**2.** Sterilise the isolation area (air and bench) with the 75% ethanol sprayer; make sure a mist is emitted. Move the equipment back in afterwards.

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**3.** Before plating, register the **primary plate** in LIMS: `/lims-register` Stage 2a (one plate per umbrella specimen; media `MEA`, method `Direct isolation`, number of plates). Write the plate ID, media, date and your name on the **bottom** of each plate.

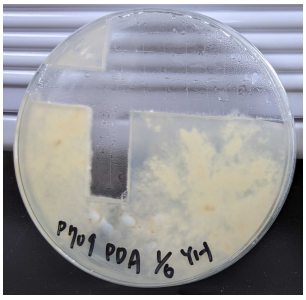



*Plate number, media, date, your name*

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**4.** Sterilise every tool before each isolation: dip the scalpel / tweezers / spreader in 95% ethanol and flame; **repeat three times**.

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**5.** Isolate microbes from the sample onto the primary plates. The method depends on the sample type (direct

plating, dilution, baiting): check the literature or ask a senior.

**6.** Seal each plate with two rounds of grafting tape, bag them (PP bag no. 8 / 10 / 12) and incubate in your incubator slot for **7–21 days**.



**7.** Check growth daily; **do not let plates overgrow**. Subculture as soon as colonies are mature.

### STAGE 2B · PICK AND SUBCULTURE ISOLATES

**8.** Under the dissecting microscope choose the target colonies. Mark each on the plate bottom with its future **secondary plate number**. Two morphotypes on one primary plate = **two** secondary numbers.

**9.** Subculture each marked colony (single colony or hyphal tip) onto a labelled PDA plate following the Stage 2a steps. These are the **secondary cultures**; they do not have a Culture# yet. Keep the primary plates until every secondary plate is clean; if a secondary plate is contaminated, redo it from the primary.

### PRESERVATION (STAGE 2B → VIALS)

**10.** Obtain the **Culture#** first: register the isolates with `/lims-register` Stage 2b, **one row per isolate** (source = the Stage-2a plate; media `PDA`; strain code → `morph_description`; provenance → notes). Review the preview, then confirm: LIMS mints one **Culture#** per isolate, in row order, and creates the water-vial preps. Registered rows cannot be deleted, so check the list first.

**11.** After **7–21 days** the secondary cultures are ready. Per **Culture#** prepare **2 water vials + 2 glycerol vials** and **one 2 mL lysis tube** with beads. Label every cap with the Culture# (black marker / technical pen).

**12.** UV the laminar flow for **15 min**; ethanol-sterilise everything going in; light the alcohol lamp. Tools as for isolation, but a **scalpel** instead of tweezers and spreader.

**13.** Flame the scalpel before every cut. Slice agar from one colony into **one water vial + one glycerol vial**; the remaining agar goes into the lysis tube for DNA extraction (Stage 3a, *DNA extraction v9.0*).

**14.** Store: water vials straight into their freezing box (steel cabinet). Glycerol vials in a freezing container at **−20 °C for 1 day**, then into the **−80 °C** collection (outside N927). Record vial positions in LIMS.

**Note** The lysis tube is the input for Stage 3a: register the extraction as a draft, quantify on the Qubit, and only then mint the MolWork#.

**Materials**

| # | Material |
|---|---|
| 1 | Media plates: MEA for primary plates, PDA for secondary plates (see *Fungal culture media*) |
| 2 | Scalpel, tweezers, spreader; 95% ethanol cooling bottle; alcohol lamp; 75% ethanol sprayer |
| 3 | Grafting tape; PP bags no. 8 / 10 / 12 (8 / 24–32 / 48–56 plates) |
| 4 | Dissecting microscope |
| 5 | Preservation vials: water vials ×2 and glycerol vials ×2 per Culture#; 2 mL lysis tube with beads (see *DNA extraction*) |
| 6 | Black marker or technical pen; freezing boxes (water vials: steel cabinet by Prof. Huang's office; glycerol vials: −80 °C freezer outside N927) |